Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/12/2005
PI Name:	Muirhead, Dean Ph.D.		
Project Title:	Development, Testing and Verification of an Anoxic Biological Water Processor		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-3070
Organization Name:	NASA Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	05/15/2003	**End Date:**	08/01/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Campbell, Melissa		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The maximum hydraulic and organic loading limits of attached growth, biological wastewater treatment systems have not been well defined, particularly for gravity independent systems designed for space applications. In addition, the useful lifetime of these biofilm systems for a given crew size, mission duration, and wastewater stream is unknown. The main goal of this project is to answer the question: What is the minimum size of the anoxic reactor required to treat a given crew's wastewater for a specified period of time? To answer this question, organic loadings will be varied by increasing hydraulic loading of a constant influent organic concentration representing urine, urine flush, humidity condensate, and hygiene water. The limitations of the biological system will be determined based on system performance, effluent quality, and measured organic and nitrogen removal efficiencies. The results of this study will provide design guidelines for sizing anoxic bioreactors to remove organic carbon and nitrogen from wastewaters containing personal cleansing agents, humidity condensate, and urine. 1. Determine the maximum hydraulic loading rate		

Task Description:	to the pressurized predenitrification BWP system. 2. Determine the steady-state organic carbon and total nitrogen removed by the anoxic denitrification bioreactors operating at the maximum hydraulic loading rate. 3. Estimate the rate of biomass accumulation and the effects of this accumulation on accessible surface area, working volume, pressure drop, and treatment performance of the denitrification bioreactor. 4. Determine the lifetime of the anoxic denitrification bioreactor operating at the maximum hydraulic loading rate. 5. Estimate the minimum size of the anoxic denitrification bioreactor to treat EPB wastewater for a given crew size and mission duration. BWPs are currently at a Test Readiness Level, TRL 4, meaning they have been validated in laboratory environments. The next step is to optimize and integrate the BWPs with other life support sub-systems. In order to advance the BWP TRL level beyond 4, some questions about the reliability, size, and long-term performance of BWPs must be answered.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Reuse of water on Earth is becoming more important as populations increase, increasing the demand for a limited supply of fresh water. Reusing wastewater is becoming more acceptable and economically feasible. This project develops water reuse technology that has countless applications on Earth. The biological water processor requires little energy and space.
Task Progress:	The test stand is completed and in operation. The nitrification bioreactor has been inoculated and is currently treating 1 L per day of Nitrification ersatz containing 700 mg of total ammonia nitrogen per liter. The hyraulic loading on the nitrification bioreactor is being increased once steady state conditions are reached to establish a nitrifying microbial population prior to coupling with the denitrification bioreactors. The denitrification bioreactors are currently undergoing hydraulic tracer (bromide) studies as a function of flow rate.
Bibliography Type:	Description: (Last Updated: 02/27/2004)